Safer Ground, Tighter Markets:

A Dynamic Two-Market Model of Housing Adjustment under Risk Rating 2.0

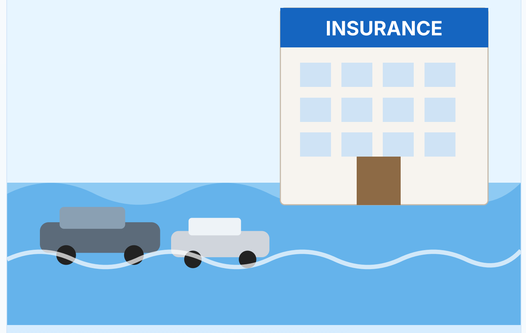
Kosar Hosseini · Katherine Yoder Zipp · Scott Colby · Zhengqi Liu
Pennsylvania State University · SUNY Morrisville

1 Motivation

WHY THIS MATTERS

Flooding is a common and costly natural hazard in the United States. Historically subsidized or weakly risk-based flood insurance lowered the true cost of high-risk housing and encouraged development in exposed areas.

Risk Rating 2.0 (RR2) was introduced in 2023 to align NFIP premiums more closely with property-level flood risk.



Key question: Does a policy targeted at floodplain homes affect only those areas, or does it also spill over into safer areas, and tighter nearby markets?

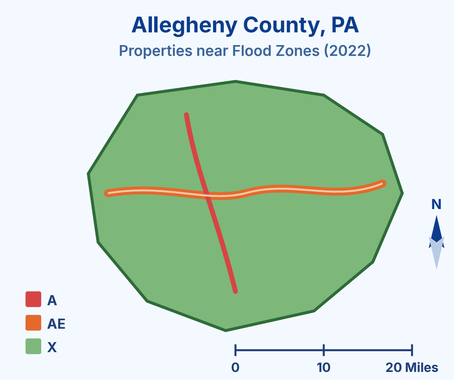
2 Research Question & Case Study

How does risk-based insurance pricing affect prices and housing stocks both inside and outside the floodplain?

Case study: Allegheny County, PA

Data: CoreLogic parcel-level residential data + FEMA National Flood Hazard Layer

Baseline year: 2022



3 Economic Mechanism

HOW RISK PRICING AFFECTS HOUSING MARKETS

Inside floodplain (higher insurance costs)

RR2 premium	↑
ownership cost	↑
willingness to pay	↓
price	↓
construction / redevelopment	↓
long-run stock	↓

DISPLACEMENT AND SPILLOVER



Outside floodplain (safer areas)

demand shifts toward safer housing	
price	↑
construction	↑
long-run stock	↑

4 Dynamic Two-Market Model

HOUSING PRICES AND SUPPLY OVER TIME

Demand side (inverse demand function)

$$R(S_N + S_F) = \frac{\alpha(W - \beta_c)}{(S_N + S_F) - \beta_s(1 - \alpha)}$$

Supply side (housing supply function)

$$Y_i(P_i) = a_i P_i^{\varepsilon}, \quad i \in \{N, F\}$$

Calibration equations

$$\varepsilon_{s,w} = \frac{\alpha W}{R^{eq}\beta_s(1 - \alpha) + \alpha(W - \beta_s)}$$

$$\theta = \alpha + \frac{R^{eq}\beta_s(1 - \alpha) + \alpha\beta_c}{W}$$

$$a_i = \frac{\delta S_i^{eq}}{(P)^{\varepsilon}}, \quad i \in \{N, F\}$$

Dynamic system (N: non-floodplain, F: floodplain)

$$\dot{P}_N(t) = \psi_N P_N(t) - R(S_N(t) + S_F(t))$$
$$\dot{P}_F(t) = \psi_F(m)\, P_F(t) - R(S_N(t) + S_F(t))$$

$$\dot{S}_N(t) = a_N P_N(t)^{\varepsilon} - \delta S_N(t)$$
$$\dot{S}_F(t) = a_F P_F(t)^{\varepsilon} - \delta S_F(t)$$

$$\psi_F(m) = \psi_F^0 + f\, m > \psi_N$$

5 Allegheny County Calibration (2022)

	Non-FP	FP
Housing stock	379,126	4,036
Median price	$218,000	$110,000
Market share	98.95%	1.05%

Calibration inputs

Median income	$76,000
Supply elasticity	1.20
User cost rate	0.059
Housing expenditure share	0.26
Income elasticity of housing demand	0.67

Main simulation

82% increase

in the flood-insurance premium component (relative to baseline).

6 Main Results

Allegheny County — 82% premium increase

Outcome	Floodplain market	Non-floodplain market
Short-run price change	↓ −29.0%	↑ +0.05%
Long-run price change	↓ −29.0%	↑ +0.15%
Long-run stock change	↓ −33.0%	↑ +0.18%
Total regional stock change	↓ −0.17%	

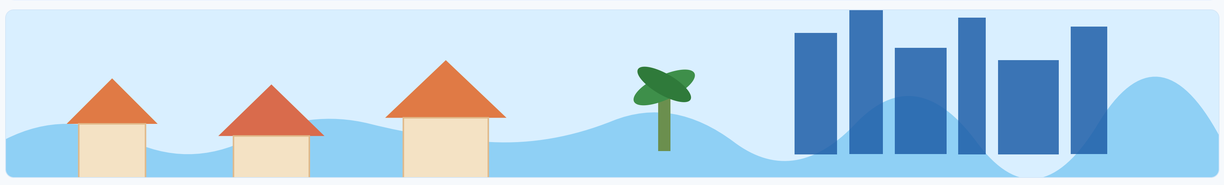
7 Policy Implications

FROM RISK PRICING TO MORE RESILIENT HOUSING MARKETS

RR2 improves risk pricing and can discourage excessive exposure in high-risk areas.

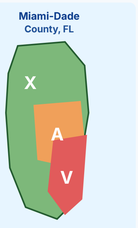
Spillovers affect safer markets too. Complementary housing-supply and affordability policies may be needed in safer areas.

Place-based climate policy should consider regional spillovers, not only direct treatment effects.



8 Miami-Dade extension — 85% premium increase

50.48% of properties are within SFHA.



- X — outside SFHA (safer market)
- A — inland (high-hazard)
- V — coastal (high-hazard)

	Percentage changes		
	Short run (outside)	A (inland)	V (coastal)
Short-run price change	+1.04%	−6.79%	+17.94%
Long-run price change	+2.60%	−5.25%	−16.68%
Long-run stock change	+3.13%	−6.26%	−19.67%

Approx. total regional supply change: −1.69%

9 Key Modeling Assumptions & Future Extensions

Baseline Assumption	Alternative to Explore	Future Research
Perfect substitution. Common rental value based on total housing stock.	Imperfect spatial substitution (CES).	Examine neighborhood choices and migration patterns.
Common supply elasticity. Same elasticity in both markets.	Market-specific supply elasticities.	Use developable land and zoning constraints to inform calibration.
Mobility not explicitly modeled. No household relocation decisions.	Moving costs and mobility frictions.	Calibrate relocation frictions using empirical literature and micro-data.

Next steps: Assess how spatial substitution, supply constraints, and mobility frictions affect housing-market spillovers.

